

RESEARCH ARTICLE

Consumer Response to the Use of AI in Value Co-Creation in Online Communities—The Case of *Nexus Mods*

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ABSTRACT

The article draws on the theoretical framework of behavioral reasoning theory and online communities literature to examine how the inclusion of artificial intelligence (AI) is perceived by consumers when it is used during value co-creation occurring outside of typical market logic. The study specifically focuses on a community built around the *Nexus Mods* platform, where consumers create and share mods (modifications) for video games, without involvement or governance from the games' developers. Adopting a mixed-method approach, involving a quantitative study and netnographic research, the article shows that despite displaying anti-AI bias, consumers are still willing to adopt AI mods. It also explores reasons for adoption (high quality, enriched experiences, more choice, novelty of AI, consumer empowerment, consumer resistance) and against adoption (low quality, "lack of soul," negative social impact, ethical considerations, legal concerns, and lack of transparency). Finally, the article demonstrates how the context of the online community makes consumers more accepting of the use of AI, as it no longer serves the purpose of profit maximization by companies. Simultaneously, the inherent issues of AI, such as being strongly rooted in the exploitation of intellectual property and labor of artists and other actors, still prevent the technology from being accepted unanimously and unequivocally.

1 | Introduction

AI is widely perceived as a highly disruptive technology, leading to the transformation of businesses across different sectors (Cheng et al. 2022). The consumers' adoption of AI and their attitudes towards this technology have been studied in the context of multiple different industries, including retail (de Bellis and Venkataramani Johar 2020), healthcare (Gaczek et al. 2023; Kumar et al. 2024), automobile (Huang and Qian 2021), and banking (Lee and Chen 2022). The results paint a complicated picture. On the one hand, AI-enhanced products and services seem to excel in a number of tasks, such as consumer service (Tran et al. 2021), medical diagnosis (Gaczek et al. 2023; Kumar et al. 2024), and idea evaluation (Siemon 2023). Moreover, some AI solutions such as ChatGPT and MidJourney have been

widely adopted by consumers (Kumar et al. 2025). On the other hand, the general consumer adoption in other sectors, such as healthcare (Longoni et al. 2019) and autonomous cars (Huang and Qian 2021) remains limited.

The literature also points to AI's numerous negative aspects, such as worker displacement (Brynolfsson 2023; Modliński et al. 2023; Robert et al. 2020), relying on unethically obtained data (Lnu 2023; Morreale et al. 2023) or reproduction of existing social inequalities (Acemoglu and Restrepo 2020; Baradaran 2024). Consequently, researchers have noticed consumers displaying an anti-AI bias (Chamberlain et al. 2018; Horton et al. 2023) or human favoritism (Zhang and Gosline 2023), meaning that the mere disclosure of AI usage by the company (AI transparency) can negatively impact consumer adoption (Luo et al. 2019).

Despite the topic of AI consumer adoption gaining a lot of attention in recent years, there still remain some unexplored gaps in the literature. First, the existing studies predominantly use quantitative research methods, resulting in a lack of qualitative, exploratory research regarding AI consumer adoption (Jain, Wadhvani, et al. 2024). In particular, there are no studies using netnography. Supplementing quantitative research with qualitative data is important, as it provides a more complete picture of the studied phenomenon (Venkatesh et al. 2013).

Moreover, there is a lack of studies that explore the consumer adoption of AI and its ties to AI transparency in the context of the video game market. This is an important oversight, as AI can potentially be used for many different tasks in the video game development process, such as writing, coding, and asset creation. Simultaneously, its use in this market also seems to stir a lot of controversy among consumers (Hernandez 2022; Price 2023). Both of these factors make the video game industry an interesting environment for examining consumer attitudes toward AI and its adoption.

Last but not least, the existing literature focuses on consumer adoption in the usual market context, where companies incorporate AI in their commercial offering. AI has been mostly studied as a tool used by companies to create their value proposition (see, e.g., Gaczek et al. 2023; Nagy and Hajdú 2021; Schepman and Rodway 2020; Wang et al. 2023) or to replace human employees in the process of value co-creation, through, for example, AI chatbots (Castillo et al. 2021; Jiang et al. 2022; Luo et al. 2019; Shin et al. 2023) or AI voice assistants (Anayat et al. 2023; Kamoopuri and Sengar 2024; Kim and Kim 2024). However, there are no studies on consumer adoption in the alternative, non-commercial context of online communities, where AI is used by consumers in the process of co-creation of value that is then appropriated by the members of the community (Jain, Sharma, et al. 2024). This is particularly important, as technology adoption is context-dependent (Tamilmani et al. 2021), and thus examining it in different contexts can shed new light on the conditions behind consumer acceptance of AI or lack thereof.

Thus, the purpose of this article is to examine consumers' adoption of AI-based products co-created in online communities. For this purpose, using the theoretical framework of behavioral reasoning theory and online community literature, the article examines the community built around a video game modding platform *Nexus Mods*. Mods are optional bonus features or re-designs of a game usually developed from the bottom up by the fan community (Klimas 2019). The article investigates consumer attitudes towards and adoption of mods that have been developed using AI. Thus, four research questions have been formulated:

RQ1. How does the use of AI in mod development impact consumer adoption?

RQ2. How does the AI transparency of AI mods impact consumer adoption?

RQ3. What are the reasons for and reasons against the consumer adoption of AI mods?

RQ4. How does the context of the online community impact consumer adoption?

The research adopts a mixed-method approach. First, quantitative research is conducted to provide answers to RQ1 and RQ2. By using the API of *Nexus Mods*, the consumer adoption of mods is examined. Initially, the consumer adoption of AI mods is compared with the adoption of non-AI mods. Then, the consumer adoption of AI mods is compared between mods characterized by high and low AI transparency. In both cases, the comparisons are based on the two-tailed Mann-Whitney *U* tests.

Then, to address RQ3 and RQ4, netnographic research is carried out, involving the analysis of the textual data from forum posts, comments, and mod descriptions discussing AI in the context of modding. The netnography, as a qualitative method, is used “to foster an understanding that encompasses cultural characteristics such as meanings, identities, hierarchies and rituals” (Kozinets 2023, 106), thus helping explain the “why” behind the quantitative results and providing deeper insights into the consumer adoption of AI mods co-created by the online community.

The study's findings demonstrate that consumers are willing to adopt AI mods, despite showing anti-AI bias. Moreover, the research identifies multiple reasons for the adoption of AI mods (high quality, enriched experiences, more choice, novelty of AI, consumer empowerment, and consumer resistance) and against adoption (low quality, “lack of soul”, negative social impact, ethical considerations, legal concerns, and lack of transparency). Last, but not least, the results show that some consumers adjust their attitudes towards AI depending on the context, being more accepting of it when used for value co-creation within an online community, rather than by companies in pursuit of profit. For managers, this means that while AI is overall a controversial technology, consumers' negative attitudes are tied more to the way it is applied by companies rather than to the technology itself.

This article is organized as follows: first, the consumer adoption literature is reviewed. Next, the theoretical background for the study is established, with the subjects of AI, AI transparency, online communities, and value co-creation being discussed. Next, the methodology employed in the study is presented. The following section contains the results of the conducted studies. The article ends with a discussion of the results, as well as their theoretical and practical implications.

2 | Literature Review

There are many models and theories explaining the adoption of new technologies and products based on them, such as the technology acceptance model (Davis 1989) or the [unified theory of acceptance and use of technology](#) (Venkatesh et al. 2003). However, they tend to focus on positive factors influencing consumers' adoption behavior, while ignoring the ones that have a negative impact on adoption (Li and Wang 2024), which limits their usefulness when examining controversial technologies such as AI. Thus, for examining the attitudes of consumers adopting AI technologies, as well as products and services developed around them, other theories, such as Cognitive Reasoning

Theory and Cue Utilization Theory, are better suited (Jain, Wadhvani, et al. 2024).

Behavioral Reasoning Theory (Westaby 2005) serves as the main theoretical framework for the study, as it pays explicit attention to reasons (both positive and negative) behind actions undertaken by consumers. It posits that reasons for and reasons against a given behavior are crucial in the decision-making process since they allow consumers to rationalize and justify their behavior (Hajiheydari et al. 2021; Lalicic and Weismayer 2021). Thus, they are context-specific antecedents of the global variables such as attitude toward the behavior, subjective norms, and perceived control, employed in earlier models, such as the theory of planned behavior (Fishbein and Ajzen 1975) and the theory of reasoned action (Ajzen 1991), upon which BRT builds upon. These contradictory reasons, such as advantages and disadvantages, risks and benefits, enablers and inhibitors, are crucial for explaining adoption intention and adoption itself (Anayat et al. 2023).

BRT has been widely employed in research on consumer adoption behaviors (e.g., Claudy et al. 2015; Gupta and Arora 2017; Shukla et al. 2024). There are also a number of publications that use BRT to specifically examine consumer adoption of AI-related services and products. For instance, BRT has been employed as a framework to examine reasons behind the adoption of AI-based voice assistants (Acikgoz et al. 2023; Anayat et al. 2023), chatbots (Lalicic and Weismayer 2021), and autonomous cars (Huang and Qian 2021).

Crucially, the vast majority of these studies implement quantitative methodologies. Thus, as noted by Jain, Sharma, et al. (2024) there is a lack of qualitative, exploratory studies regarding AI consumer adoption. There are also no studies using netnography in particular. This is one of the gaps that this study aims to fill. Moreover, there are no studies that explore AI adoption in the context of the video game market, which is addressed by this study. Last, but not least, the current literature focuses on consumer adoption specifically in the context of commercial products and services sold for profit, thus not accounting for different, non-commercial contexts, such as online communities. This is the third research gap that this study aims to address.

3 | Theoretical Background

3.1 | AI

AI was first conceptualized by McCarthy, who defined it as machines and computer programs capable of displaying human-like intelligence (Cordeschi 2007). Since then, this concept has gained increased attention from both scholars and the general public due to the introduction and proliferation of generative AI platforms, chief among them ChatGPT (Lim, Gunasekara, et al. 2023). Generative AI encompasses a myriad of different solutions developed through machine learning that allow users to generate output (such as text, images, sounds, and videos) based on prompts (Sætra 2023). Therefore, it has been seen by some as a potentially revolutionary piece of technology (Fui-Hoon Nah et al. 2023) that can empower consumers and enrich their experiences (Han and Broniarczyk 2022).

While consumers appreciate its novelty (Sung et al. 2021), its acceptance and adoption also depend on its quality (Jain, Wadhvani, et al. 2024; Sung et al. 2021), consumer's overall attitude towards this technology (Moriuchi et al. 2021) and the perceived ethical and societal ramifications of its use (Gerlich 2023). Moreover, some consumers still hold a strong preference towards human creativity over that of an algorithm, especially in the context of art-related products and services (Bellaiche et al. 2023; Zhou and Kawabata 2023).

Consumer adoption and attitudes regarding AI have been studied in the context of various sectors and industries, such as retail (de Bellis and Venkataramani Johar 2020), healthcare (Gaczek et al. 2023; Kumar et al. 2024), fashion (Guo et al. 2023), marketing (Spais et al. 2024), automobile (Huang and Qian 2021), and banking (Lee and Chen 2022). However, there is a general lack of studies regarding the application and adoption of AI in the video game market. Thus, the article strengthens the existing literature by providing insights regarding this yet unexplored industry.

3.2 | AI Transparency

The importance of transparency in business has been examined by many researchers in various contexts, such as brand-building and advertising (Guèvremont 2018; Hainneville et al. 2023), product labeling (Cho et al. 2024) and algorithms (Lehmann et al. 2022). It can help establish consumers' trust, a feeling of safety (Qin and Brown 2008; Wang and Benbasat 2007), and authenticity (Cho et al. 2024; Guèvremont 2018), thus positively influencing purchase intentions (Veltri et al. 2023).

AI transparency involves the disclosure of AI-generated content and allows consumers to make informed decisions (Mogaji and Jain 2024). However, similar to algorithm transparency (Lehmann et al. 2022) it can also have adverse effects. For example, AI transparency was found to cause a decrease in trust from consumers (Zarifis et al. 2021), and a steep reduction in call length in customer support (Luo et al. 2019).

The study builds on the existing literature by examining the impact of AI transparency on consumer adoption in a different context, shifting the focus from company-developed offerings to those developed within an online community.

3.3 | Value Co-Creation in Online Communities

Online communities are developed through continuous interactions among actors who share similar interests and practices, as well as use similar tools (Dahlander and Frederiksen 2012). Oftentimes, they are created and controlled by companies, who seek to involve their consumers in the value co-creation process and manage it from above (Yeritsian 2021). For that purpose, they provide consumers with tools to allow customization of the products and thus help shape their consumer experience (Frow et al. 2015; Sheth 2017; Vargo and Lusch 2004). Such value co-creation necessitates mental processing of the offering (Vargo and Lusch 2008), but can also involve more objective modifications of the value proposition, which occurs through means

such as co-conception of ideas, co-design, co-production, or co-maintenance (Sheth and Uslay 2007, 305).

However, online communities can also exist and act independently from the company, rarely if ever interacting with it, while still being impacted by it and influencing it in return (Cyron et al. 2024; Fisher 2019). Even online communities built around a single brand, can rise spontaneously, through “participation from below,” and act independently from the brand itself (Yeritsian 2021) since due to the shared resources of the online community and the proliferation of digital tools that make editing products easier, value co-creation processes in online communities become less and less dependent on the actual companies (Wieczerzycki and Deszczyński 2022). In many cases, the company's involvement becomes limited to providing the initial product, which is then individually or collectively processed by consumers, with the resulting value being then appropriated by other consumers (Cova and White 2010). Such value co-creation in online communities sometimes occurs even without the knowledge of the company, or against its explicit will. However, as it can ultimately improve on the initial value proposition, it can still be perceived as free digital labor for the companies (Sugihartati 2020). Thus, while such value co-creation can be empowering to the consumer (Füller et al. 2009), it can also be exploitative, leading to problems such as unpaid work (Cova et al. 2015; Daros 2022).

The studies on such independent online communities are limited. Cova and White (2010) carried out netnographic research on Warhammer tabletop wargaming and CouchSurfing communities (which they refer to as counter- and alter-brand communities, respectively). Yeritsian (2021), also using a netnographic approach, examines the dialectic between brand-directed and consumer-directed attempts at shaping the storylines in wrestling. Both these studies focus on the communities' symbiotic and adversarial relationships with the brands they are built around. This article further develops these works by employing a mixed-method methodology, which involves quantitative research. This sheds more light on the actual consumer adoption of the solutions developed within the online community, and how this adoption is tied to AI technology. Moreover, it shows how consumers belonging to a counter-brand community perceive AI as a tool for resisting companies, thus rationalizing its use.

4 | Hypothesis Development

4.1 | Impact of AI Usage in Video Game Modding on Consumer Adoption of Mods

The consumer adoption of AI technologies has been mixed. On the one hand, some general-purpose services, such as chatGPT or MidJourney, have managed to gather impressive numbers of users (Kumar et al. 2025). Similarly, voice assistants have also become popular with consumers (Poushneh 2021).

However, for many other, more specialized applications, factors such as mistrust and apprehension have limited consumer adoption (Hasan et al. 2021). Moreover, as the technology is still

relatively new, many consumers remain on the lower levels in the diffusion of innovation ladder (Jain, Wadhvani, et al. 2024), which suggests that cases like chatGPT and MidJourney might be outliers. In many sectors, such as healthcare (Longoni et al. 2019) and autonomous cars (Huang and Qian 2021), consumers remain resistant to AI technology and thus the adoption of products and services that are based on AI remains relatively small. Therefore, similar resistance can be expected in the context of video game mods. Hence, the following hypothesis is put forward:

H1. *AI use in the production of video game mods negatively impacts mod consumer adoption.*

4.2 | Impact of AI Transparency in Video Game Modding on Consumer Adoption of AI Mods

Consumer adoption doesn't depend exclusively on the quality of the product/service, but also, i.e., on the consumer's attitude towards it and the technologies it is based on (Ajzen 1991; Westaby 2005). In the case of controversial technologies, such as AI, this can lead to so-called algorithm aversion (Lehmann et al. 2022). Specifically, various authors noticed consumers display an anti-AI bias (Chamberlain et al. 2018; Horton et al. 2023), or human favoritism (Zhang and Gosline 2023). Consequently, some studies show that AI transparency negatively impacts consumer adoption. For instance, according to Luo et al. (2019), consumers buy less after the disclosure of the fact that the call was made by an AI chatbot. Similarly, Tong et al. (2021) discovered that employees are less likely to learn from AI feedback after the disclosure. Thus, a similar effect of AI transparency on user adoption can be expected in the context of video game mods. Based on this, the following hypothesis is proposed:

H2. *High AI transparency in video game mods negatively impacts AI mod consumer adoption.*

5 | Research Design

5.1 | Research Context

The research concerns the online community built around the *Nexus Mods* modding platform. Modding, “from modifying, is the act of changing a game, usually through computer programming, with software tools that are not part of the game” (Poor 2014, 1250). It involves creating optional bonus features or modifications of a game and is usually organized from the bottom up by the online fan community (Klimas 2019), and as such, constitutes an act of value co-creation. *Nexus Mods* is one of the largest existing modding platforms, with over 40 million registered users, and over half a million hosted mods, which have accrued more than 10 billion downloads to date (Nexus Mods, n.d.). The mods published on the platform range from simple modifications focused on addressing a specific issue with the game (e.g., fixing a bug, increasing/decreasing difficulty level, improving texture resolution, adding new items, removing triggering/fear-inducing elements, etc.), to comprehensive overhauls, drastically changing visual style

of the modded game, its gameplay systems and mechanics, introducing new technologies (e.g., real-time cloth physics simulation, resolution upscaling algorithms, or physical collision algorithms) or creating altogether new worlds, adventures and narratives using the modded game engine. Each mod receives its own description page, screenshot gallery, comment section, version history, and so forth. The users can signify their appreciation of the mod by endorsing it, which functions similarly to the like button found in more traditional social media platforms. The mods hosted on *Nexus Mods* are free. However, some of the authors do try to monetize their works using crowdfunding services such as Patreon.

Since April 27th, 2023, *Nexus Mods* has an established policy regarding the use of AI in mods hosted on the platform. The AI-generated content is generally allowed; however, in the case when a third party makes a substantiated complaint about their rights being infringed, the AI mod in question will be removed. Thus, the mod creators are encouraged to avoid using such tools without the explicit permission of the parties holding the rights that might be infringed by AI-generated content (AI-Generated Content in Modding at Nexus Mods and Community 2023). This stance has generated a heated discussion between consumers using the platform, displaying the lack of consensus on the matter of AI among the users.

5.2 | Methods

The article involves a mixed-method approach. The first study is quantitative in nature, which allows researchers to test concepts and relationships defined a priori (Fischer and Guzel 2023) and thus answer narrow questions in an unbiased, objective manner (Haradhan 2020). This study begins by comparing consumer adoption between AI mods and non-AI mods. Then the adoption of AI mods characterized by high AI transparency and those with low AI transparency is compared. The data was processed using SPSS statistics. Due to the non-normal distribution of the data and a large number of outliers, two-tailed Mann–Whitney *U* tests were used to compare groups. For all tests, statistical significance required a two-tailed $p < 0.05$. For the same reason, the performance of the mods has been evaluated based on the median and the 25 and 75 percentiles rather than mean values.

The second study employs a netnographic approach (Kozinets 2015) to provide deeper insights into consumer attitudes towards AI mods on the *Nexus Mods* platform. This method entails observing, collecting, and interpreting consumers' "online traces" (Kozinets 2019). This choice was made because "netnography complements the phenomenological, social, interactive and networked orientation of most modern service experience co-creation research, such as service-dominant logic and consumer culture theory-based service research" (Kozinets 2023, 106) and has been used in many studies regarding online communities (i.e., Dametto and Bonet-Marti 2024; Hewer and Brownlie 2013; Jones 2015; Zestanakis 2020).

Using this combination of quantitative and qualitative methods allowed for combining the strengths of both approaches,

providing a better understanding of the research problem than either approach alone (Molina-Azorin 2016). Specifically, using netnography made it possible to expand on the results of the quantitative research, providing a more complete and complementary picture of the studied phenomenon (Venkatesh et al. 2013).

5.3 | Data Collection and Processing

5.3.1 | Study 1—Quantitative Analysis

The data for the quantitative study was gathered using the API of the *Nexus Mods* website and a script written in Python. Since the API does not allow the retrieval of data on the latest mods for all the games, and can only provide the latest mods for a specific game, five games were chosen. The choice was made based on the total number of downloads for all the games' mods. The resulting list of titles included: *The Elder Scrolls V: Skyrim Special Edition*, *The Elder Scrolls V: Skyrim*, *Fallout 4*, *Fallout 3: New Vegas*, and *Stardew Valley*. Since the first two titles are essentially the same game (with the special edition being a later re-release) and share a significant amount of mods, the decision was made to omit the original version so as to avoid skewing the results by over-representing a single game. Instead, the next game meeting the established criteria—*The Elder Scrolls IV: Oblivion*—was added to the list. For all the games included, the data regarding all the mods published from the 30th of May 2023 to the 31st of October 2023 was collected. In order to adhere to *Nexus Mods* API's terms of service and not disturb the functioning of the platform, the daily and hourly request limits were strictly followed. As a result of the process, the data on 7677 mods has been retrieved. Thus collected information included: the title of the mod, summary, full description, title of the modded game, creation date, category, total number of unique downloads, and total number of endorsements.

To separate the mods that were created with the help of AI from those that did not use this technology, the titles, summaries, and full descriptions of each mod have been checked for the inclusion of keywords related to AI and AI-based software used in modding. The list included the following keywords: ai-, ai generated, by ai, artificial intelligence, machine generated, machine-generated, ChatGPT, Chat GPT, xVASynth, ElevenLabs, Eleven Labs. 142 mods including these keywords have been identified. The titles, summaries, and full descriptions were then examined manually to confirm that they do use AI in some manner. Based on this, 38 mods have been excluded from the group. Ultimately, 94 mods using AI have been identified in the sample, with 7583 remaining mods being created with no help from AI.

5.3.2 | Study 2—Netnographic Research

For the netnography, the data collection stage involved two phases (Kozinets 2019). The first, immersive phase involved the researcher immersing themselves in the studied environment to understand the culture built around video game modding. Then, in the investigative phase, specific data relevant to the research questions was gathered. All data possessed text format and was collected from the *Nexus Mods* platform and included:

- Posts from Nexus mods forums (gathered using the search feature of the forums, using the search query: AI OR xVASynth OR ElevenLabs).
- Mod descriptions (of mods mentioning AI in the description; this data was already obtained for Study 1).
- User comments under Nexus Mods' AI-Generated Content policy.

Before the data from all the sources were combined, each piece was examined manually to filter out data irrelevant to the research subject (e.g., discussing different types/meanings of AI, making no statements on AI, etc.). In total, 667 posts, descriptions, and comments have been included in the study.

After the data was obtained, the Integration stage began (Kozinets 2023), during which data was grouped and interpreted. During the coding, the data was first divided based on the general sentiment towards AI mods (positive, negative, ambivalent, and no attitude). Then, an in-depth analysis of the reasoning behind the sentiment was carried out. The coding employed a constant comparison method, where data, codes, and themes are continuously re-examined and compared, allowing for new codes and themes to emerge from the process (Williams and Moser 2019). Thus, both the deductive and inductive coding were used (Kozinets 2010). Deductive coding was employed for the general sentiment codes (as the list of codes was predetermined before the data analysis). For the in-depth attitude analysis, as it was nigh impossible to compile the complete list of codes that would cover all the potential consumer attitudes, the study opted for using inductive codes, which emerge during the process of data analysis. Such codes “feel closer to the empirical phenomenon because they arise from contact with it” (Kozinets 2010, 339).

All the coding was done manually using MS Excel. While the automated, lexicon-based approach (e.g., Hipson et al. 2021; Mohammad and Turney 2010) was considered for the sentiment analysis, due to the specific language used by the Nexus Mods users, as well as the prevalence of sarcastic statements, the usefulness of such an approach would be limited (Alqahtani et al. 2023; Sharma et al. 2024).

After the initial coding, the higher-order constructs were developed by combining the individual codes into broader themes in the community's discourse regarding AI in modding. This process was carried out manually, “by looking across instances and contexts for meaningful similarities and other conceptual threads that connect individual codes” (Kozinets 2010, 343). Thus, created themes establish relationships between the codes and derive meaning from them (Chakraborty et al. 2024).

In netnographic research, ethical concerns are of crucial concern. In particular, “representation of research participants should be ethically responsible” (Kozinets 2023, 108). For this reason, neither nicknames of authors, nor direct quotes are provided in the Results section, with the paraphrases being used instead. This was done due to the risk of participants being identified using search engines and harassed, due to AI being a contentious issue.

5.4 | Measurement

The consumer adoption of mods was measured based on two values retrieved from the Nexus Mods Platform through its API: (1) the mod's total number of unique downloads (i.e., only a single download per user was included) and (2) the total number of endorsements. The former was included as a common way to measure software adoption (Shengli and Fan 2019). Endorsements, on the other hand, are the platform's version of likes. This item was used as it expresses consumer satisfaction, which predicts more long-term usage (Yousaf et al. 2021).

AI transparency was determined based on the analysis of the titles, summaries, and full descriptions of the mods. In particular, the placement of the AI disclosure was examined (Heo et al. 2022). High transparency involved the inclusion of the AI disclosure either in the title of the mod or its summary (both of which are immediately visible in the Nexus Mods search result pages). Conversely, mods characterized by low AI transparency only included the AI disclosure in the full description (which is only available on the mod's dedicated subpage and tends to be lengthy and packed with technical details, which in turn lowers the visibility of the disclosure).

6 | Results

6.1 | Study 1—Quantitative Analysis

To examine the consumer adoption of AI mods and test H1, first, AI mods and non-AI mods were compared in terms of the number of unique downloads. The Mann-Whitney comparison between the two samples revealed a significant difference ($Z = -5.048$, $p < 0.001$). The descriptive statistics for unique downloads have been presented in Table 1.

As demonstrated, the median of unique downloads is greater for the AI-based mods (984) than for the non-AI mods (285), thus not supporting H1. Similarly, the 25 and 75 percentiles are also substantially higher for AI mods. However, there is also a number of greatly performing outliers in the non-ai mod sample,

TABLE 1 | Number of unique downloads in non-AI mods and AI mods.

		Unique downloads				
		Median	Percentile 25	Percentile 75	Minimum	Maximum
Mod type	Non-AI mods	285	105	1029	1	151,770
	AI mods	984	223	4425	12	18,236

Source: Own elaboration.

TABLE 2 | Number of endorsements in non-AI mods and AI mods.

		Endorsements				
		Median	Percentile 25	Percentile 75	Minimum	Maximum
Mod type	Non-AI mods	15	5	43	0	4462
	AI mods	44	13	151	0	458

Source: Own elaboration.

TABLE 3 | Number of unique downloads depending on AI transparency.

		Unique downloads				
		Median	Percentile 25	Percentile 75	Minimum	Maximum
AI transparency	Low	1926	387	4565	22	11,523
	High	308	136	1572	12	18,236

Source: Own elaboration.

TABLE 4 | Number of endorsements depending on AI transparency.

		Endorsements				
		Median	Percentile 25	Percentile 75	Minimum	Maximum
AI transparency	Low	72	17	201	0	372
	High	22	9	63	1	458

Source: Own elaboration.

hence the maximum value of unique downloads is substantially higher than in the case of AI mods.

Next, the number of endorsements between the two groups was compared. In this case, the Mann–Whitney test also revealed a significant difference ($Z = -5.514$, $p < 0.001$). The descriptive statistics for endorsements have been presented in Table 2.

Once again, the median of the number of endorsements is greater for the AI-based mods (44) than for the non-AI mods (15), thus failing to support H1. As was the case for the unique downloads variable, AI mods also achieved higher values for endorsements in the 25 and 75 percentiles. In terms of maximum value, non-AI mods again performed much better due to the existence of outliers.

Next, the impact of AI transparency on AI mod adoption was examined in order to test H2. The number of unique downloads and endorsements of AI mods with high AI transparency and low AI transparency was compared. Mann–Whitney comparison revealed a significant difference regarding both the number of unique downloads ($Z = -2.432$, $p = 0.015$) and endorsements ($Z = -3.039$, $p = 0.002$). The descriptive statistics for both variables have been presented in Tables 3 and 4, respectively.

In terms of unique downloads, the mods characterized by high AI transparency performed significantly worse than the ones with low AI transparency, which supports H2.

TABLE 5 | Response of Nexus Mods users toward AI.

Response towards AI	Frequency	Percent
Positive	236	35.4
Negative	138	20.7
Ambivalent	102	15.3
Undefined	191	28.6
Total	667	100.0

Source: Own elaboration.

In terms of the number of endorsements, the mods with high AI transparency again performed worse than the ones with low AI transparency, further supporting H2.

6.2 | Study 2—Netnographic Research

In general, the AI-related discourse on the *Nexus Mods* platform features a stark polarization between consumers displaying a positive attitude towards the use of AI in modding and those who oppose it (see Table 5). Many consumers are also conflicted about the technology or present nuanced opinions. In some cases, consumers did contribute to the discourse on AI; however, it was not possible to determine their stance on the subject (e.g., due to the strictly matter-of-fact character of their statements).

As illustrated by Table 6, the platform discourse can be divided into four key themes: modding, art, ethics, and society.

6.2.1 | Theme 1—Modding

The modding theme involves discussions on how AI impacts the production and consumption of mods, as well as the *Nexus Mods* platform itself. Some consumers are generally impressed with the quality of AI creations:

Eleven labs AI stuff is pretty much indistinguishable from voice acting.

honestly AI art is great

Many consumers stress the empowerment aspect, stating that using AI arms consumers with new modding capabilities.

With these new tools, you (the unskilled individual reading this) are now more capable of creating mods than ever before. No special talent is needed, just a basic understanding of a game's modding tools and a simple grasp of how to use an AI app.

People who aren't skilled in art but are great at coming up with concepts can use AI to fill in the gaps, or creators who are already good at this can speed up or expand the scale of their mods.

Some consumers thus voice their excitement with the novel aspect of the technology and the potential influx of new mods and new possibilities to enrich their experiences.

TABLE 6 | Result analysis—codes and themes.

Codes	Themes
AI's impact on mod quality	Modding
AI is nothing new	
Creator empowerment	
Rules and limitations regarding AI	
AI should be banned on the platform	Ethics
Art theft/artist replacement	
Loss of artist agency	
Conditional acceptance/moral relativism	
AI as just a tool	
Art status of AI creations	Art
Criticism of traditional art and artists	
AI as a talent equalizer	Society
AI as a technological progress	
AI as a tool of social revolution	
Inevitability of AI	
AI as a tool of social resistance	

Source: Own elaboration.

I hope we see a lot of good new content coming from this! Keep it going!

I'm really looking forward to seeing how this makes games feel more vibrant. This is exactly why people enjoy games like D&D. To experience that dynamic, living world. It won't replace it, but it will definitely add to it.

Consumers are also excited about the potentially endless amount of new options and content that AI can generate.

The idea of creating endless variations and permutations of their work with just a keystroke? Bro, we're on the brink of something incredible.

There are also more reserved voices, claiming AI is just another tool in the toolbox, which will make some tasks easier, but the overall mod quality will still ultimately depend on the individual skill and effort of its creator.

AI as software is a tool, and we've had machine-generated voices since forever, they were just bad, and no one wanted to use them. AI still needs human input, it doesn't just come up with ideas on its own.

However, many other consumers share their concerns regarding the quality of AI mods.

AI-generated stuff always looks uncanny to me, especially when it tries to create people. There's always something off that makes me uncomfortable, and once I inevitably notice it, I can't unsee it, making it unbearable.

Currently, AI-generated voices are far from replacing real voice actors in any significant project. There's no emotion or insight, and it can't 'act', it just repeats patterns.

Some of the opponents of AI in modding, as well as consumers more ambivalent about the subject, argue for the inclusion of the new rule on *Nexus Mods*, according to which all the mods using AI should be clearly marked as such.

I believe a reasonable policy change would be to add a mandatory 'AI-Generated' tag [...] I'd prefer to be able to filter and easily identify which mods were created by a human artist with a specific style versus those made by an AI through a prompt.

Despite being a form of compromise, this solution seems to nevertheless still draw criticism from the proponents of AI, who claim that such a decision would lead to further conflict.

Sure, let's tag AI, what could ever go wrong? Witch hunts? Shaming?

6.2.2 | Theme 2—Art

Oftentimes the community discourse regarding AI in modding turns into a broader philosophical debate about the nature of art and creativity, thus constituting a separate theme.

The main subject of discussion within this theme is the question of whether AI creations should be treated as instances of art, and whether they should be valued as much as art created by humans.

Only humans can create art because it's made with the soul, not the hands or mouth. Everything else, no matter how advanced, is just a tool.

Frankly, I really dislike calling AI-generated content 'art'. To me, art is about the artist's expression. AI art is just the output of a soulless, cold algorithm.

This in turn is contested by the proponents of AI, who argue that there is no fundamental difference between human and AI-created works. This subset of the community seems to display a certain manner of formalist approach to art itself (Zangwill 2005), believing that it does not matter who has created a given artwork, nor for what purpose, in what context, etc., stressing that it is just the result that matters.

Every piece of art you've ever seen forms a new neural pathway in your brain, and guess what? That's very similar to how AI operates too [...] As long as I'm being creative and there's an intent behind what I'm doing, I'm creating art.

We just want mods, who cares if they were made by a freaking machine or a human, it doesn't matter

In the context of modding, consumers draw an analogy between AI voice imitation and the voice splicing technique, which involves extracting single words or sounds from the voice lines recorded by a particular voice actor included in the modded game and using them to create new voice lines.

Using AI to generate new voice lines from a game is like splicing them together yourself, except the AI is doing the heavy lifting.

Nobody complained when we spliced lines to create new dialogue. The only distinction now is that AI sounds better.

Similarly to theme 1, some consumers perceive AI as a source of empowerment, believing that it makes artistic expression accessible to those who lack artistic talent. This belief is sometimes paired with some resentment towards artists as people, whose talents are seen as a sort of unfair privilege, with AI being presented as an equalizer of sorts.

The greatest thing about all of this is that creativity and art are no longer restricted to a select few and can be practiced by everyone.

Artists are always whining that every new technology will end their careers. They said the same thing about cameras, radio, film, video games, synthesizers in the 80s, and the internet. The only thing AI is going to do is level the playing field and allow anyone to be an artist.

6.2.3 | Theme 3—Ethics

The discourse within the ethics theme focuses on moral considerations regarding the usage of AI in modding and tends to be very heated, with various attitudes being displayed by consumers belonging to the community. A part of the discussion is focused on the legal status of AI mods and AI creations in general:

Obviously, it's a very grey area, much like most aspects of modding.

The unauthorized use of someone else's image, likeness, voice, and sound is illegal [...] All it takes is a civil lawsuit against the mod creator

I reached out to the [voice actor association] about this issue [...] Since voice actors don't own the rights to the assets or characters used to create our models, they can't really have a say in the matter. This is a problem for both sides [...] This technology is so new that the rules are still being figured out, but at the professional level, it seems that fandom use isn't the main concern for voice actors.

The strictly ethical argument from the opponents of AI in modding often focuses on the way AI models are being trained, with consumers stressing that using tools that were made using "stolen art" is ethically wrong.

Describing it as 'AI-generated' is intentionally misleading and pretentious. It's much closer to algorithm-assisted stealing.

Everyone knows that this so-called 'AI art' is just the theft of real artists' work, using it as a database for their algorithm. The so-called 'AI' lacks any true intelligence. It can't create anything; it just copies and pastes what actual artists have imagined.

There is also a group of consumers presenting a more relativist stance toward the ethics of AI usage in modding. This is usually expressed as a variation of the belief that no tool is bad in and of

itself, and thus what should be judged instead is how the tool is being used.

I want to create a more AI-friendly (or AI-neutral at least) atmosphere here, but it's important for people to fully understand the implications, legal or otherwise, of the tools they are using so they can be used correctly and not abused to exploit others' work. Abuse itself is not unique to AI, but the ways it can be misused are unprecedented.

A broadly discussed example of unethical use of AI in modding is using AI to voice NSFW (not safe for work; e.g., sexual or violent) content, without the knowledge or consent of the imitated voice actor.

The agency loss is something that I think artists are particularly afraid of. Imagine someone using old recordings of your voice to create hateful or violent messages, or taking your likeness to make NSFW content without your consent.

It's disappointing that voice actors aren't getting more protection on this website, especially when AI is being used for voicing NSFW mods.

A different frequently used relativist argument pertains to the fact that the majority of mods are works of hobbyists, and are distributed for free, as opposed to products/services created for profit by companies. This shows the contextuality of consumer attitudes towards AI.

If people were profiting from it, I can understand why that would be problematic (and illegal). However, with free mods I don't see much of an issue.

It's not that there are no issues with AI or how it's currently being used and abused by capitalist system, but it's not like the art will be dead either. Most modders don't ask for money and only share their mods with people who use them on their private PCs in their free time. This isn't exactly making a fortune off of plagiarism.

Modding games doesn't significantly impact voice actors in the grand scheme of things. The real issue will be when video game and cartoon studios start replacing voice actors with AI. [...] Some modding teams might have budgets, but most mods are solo projects done as a hobby. I don't think mod makers have an obligation to hire people.

There's also a small subset of consumers who seem dismissive toward legal and ethical considerations, displaying a quasi-nihilistic attitude.

Why do people debate values and ethics? That stuff doesn't matter at all. What counts is whether AI can create good mods. If it can, great, more stuff to download. If not, I'll just skip it.

6.2.4 | Theme 4—Society

Consumers also acknowledge the broader context behind AI technology, considering its impact on society and how it can potentially redefine the relationship between society and the modding communities.

From this perspective, many consumers perceive AI as just another instance of technological development, which will necessarily bring about changes in the way societies operate. The word “revolution” is widely invoked, with many consumers referencing historical events where breakthrough technologies simplified and streamlined production processes, while simultaneously generating social costs.

With each tech revolution, same thing happens. Remember when driving a car was dangerous, cause the engine could explode? Or flying an airplane? Learn to embrace the new technology and see what it can do.

I believe AI can be beneficial for modding. So many possibilities. However, a downside is taking jobs from paid voice actors [...] Personally, I don't like the idea of machines taking people's jobs, so I understand the concern. But ultimately, it is what it is.

Moreover, regardless of their stance on AI, many consumers seem to see the development and expansion of AI as an inevitable process, and thus believe that any sort of resistance undertaken by the modding community is doomed to fail.

AI isn't going anywhere, regardless of our objections. It's already a part of our daily lives—Siri and Alexa are AI, and we have AI-driven cars and trucks.

Unfortunately for you, you won't be able to stop it. So, prepare to live in a world where AI increasingly becomes a crucial part of our daily lives.

A small subset of consumers view AI from an entirely different perspective. They perceive using AI as a form of resistance toward the broader society and its rules, arguing that modding has always existed within the moral and legal grey area. Thus, they believe that using AI despite it being considered legally or morally dubious is a natural consequence and extension of the modding culture and its opposition toward mainstream business.

Debating AI's integrity on a modding site is hilarious. Modding has always relied on the lack of legal oversight, and much of what we do falls into a grey

area. You can't have it both ways. Before pointing fingers at individuals, let's acknowledge that the video game industry is one of the most corrupt, unregulated industries. Gamers don't get a tangible product or service, they pay for a 'license', which is one of the most exploitative concepts in modern business [...] Where's the legal scrutiny when companies manipulate content, discontinue services, or run online lotteries?

Some consumers also try to merge the stances described above by claiming that large corporations will use AI regardless of what the wider public thinks, and they will be the ones truly influencing the economic situation of workers and artists. According to this line of thinking, modders refraining from using AI on principle will just lead to a further imbalance in creative opportunities between them and corporations. On the other hand, by using AI, modding communities could more effectively oppose these corporations by offering alternative products of their own.

If AI gets banned like some of you want, you'll soon realize that all you've done is screw over the average Joe and give all the power to big corporations, who will use this technology and make sure they never lose it through lobbying.

The 'AI equals theft' crowd doesn't get that their stance could lead to big corporations having even more control over artists and their livelihoods.

This divide between 'hardworking, human, soulful artists' and 'evil, soulless, mechanistic thieves' will only benefit those who want to solidify corporate ownership of art, giving control to those with money and titles rather than the actual artists who created it.

When has that capitalist idea of copyright ever helped any artist?

7 | Summary of Results and Interpretation

The results of study 1 and study 2 provide valuable insights regarding the consumer attitude towards the use of AI in modding and the adoption of AI mods. Combining the two, Figure 1 shows the BRT-based model of AI mod adoption, showing the relations between reasons for and against adoption, global motives (attitude, subjective norm, and perceived control), and consumer adoption. The behavioral context, involving AI transparency and the non-profit environment of an online community, was also added to the model to address the research questions.

For study 1, the lack of support for H1 shows that, while the use of AI in the modding occurring on the *Nexus Mods* platform remains relatively small (amounting to little over 1% of all mods in the studied sample), its consumer adoption is slightly higher than for non-AI mods. Such results can signify a lower consumer resistance towards AI technology in the video game industry than in other industries, such as healthcare (Longoni et al. 2019) or autonomous cars (Huang and Qian 2021).

Simultaneously, the results support H2, which provides some additional insights regarding the consumer attitude towards and adoption of AI mods. The consumer adoption of mods with high AI transparency was significantly lower than mods with low AI transparency. This result suggests that consumers have a somewhat negative attitude towards AI mods, which is largely consistent with the number of existing studies on anti-AI bias and human favoritism in consumers (Chamberlain et al. 2018; Horton et al. 2023; Zhang and Gosline 2023). Moreover, these findings are in line with prior studies claiming the negative impact of AI transparency on consumer behavior, such as the ones by Luo et al. (2019).

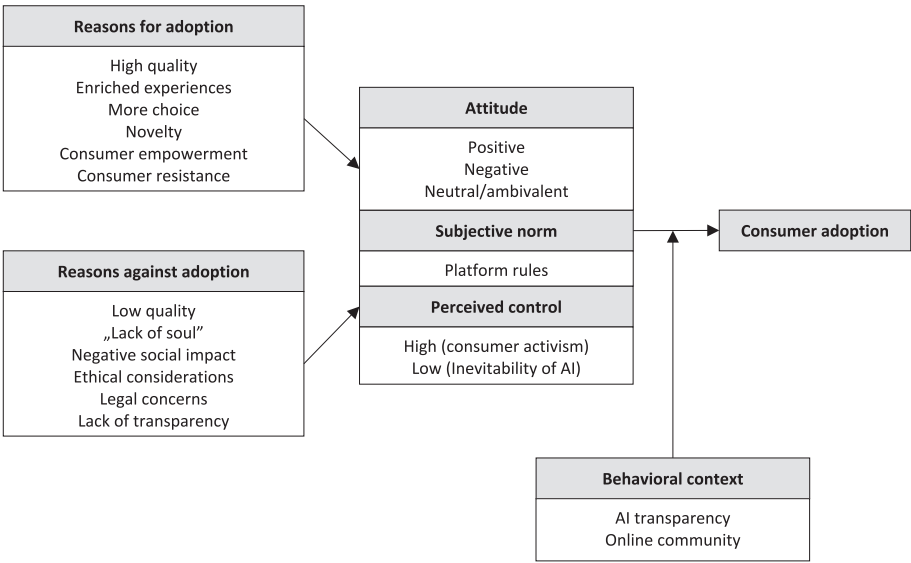


FIGURE 1 | Behavioral reasoning model of AI mod adoption.
Source: Own elaboration based on Westaby (2005).

The netnography conducted in study 2 sheds more light on the consumer attitude and reasons for/against the adoption of AI mods (summarized in Table A1). First and foremost, it shows that general attitudes towards AI vary greatly, from enthusiastic to extremely negative, which impacts the adoption (Moriuchi et al. 2021). Secondly, multiple reasons for and against the adoption of AI mods have been identified. This research showed that the quality and novelty of AI technology have an impact on consumer decisions, which is consistent with the findings of Sung et al. (2021). The empowering aspect of AI, and its ability to provide new, rich experiences, was also discussed by consumers, providing empirical evidence for the claims of Han and Broniarczyk (2022).

The study also uncovers a previously unexplored reason for adoption, namely the intention of resistance against companies and the wider society. This is likely a result of a different context, as usually the AI is used in the process of value co-creation controlled by the companies. In this case, however, it is the community that controls the process and appropriates the value co-created by using a mix of the company's and consumers' resources, making it a counter-brand community (Cova and White 2010) (although one not necessarily hostile towards the company). Similarly, the largely undiscussed reason for the adoption of AI is its capacity to provide seemingly endless content to consumers. This also seems context-specific, as modders' resources and production capacity are severely limited, which AI can counteract.

The results also show that consumers' considerations regarding the ethics of AI use, as well as its negative impact on society, also influence the acceptance of AI, confirming the claims of Gerlich (2023). Moreover, the study provides new evidence that some are unwilling to adopt AI due to its perceived lack of quality and "soullessness." This is particularly consistent with some studies conducted in the creative industry, where consumers found AI to be lacking in artistic creativity (Zhou and Kawabata 2023), unable to reflect profound human experience, and harder to engage deeply or derive meaning from (Bellaiche et al. 2023). The research also shows that in the case of some consumers, the negative attitude towards AI mods stems from the fear of lack of transparency, the consequent inability to distinguish between AI and Non-AI mods, and the resulting diminished experience with the Nexus Mods platform. Hence the insistence of some of the consumers on introducing more strict rules on the Nexus Mods platform in regard to AI content. As the existing platform rules are quite liberal, they currently don't play a large role in determining consumer behavior.

The research also provides additional insights regarding the interplay between perceived control and consumer behavior, postulated by theories such as the theory of planned behavior (Ajzen 1991) and behavioral reasoning theory (Westaby 2005). The netnography revealed that consumers differ in regard to the level of perceived control they have over the spread and adoption of AI. These perceptions were then used to rationalize their decisions, particularly in the context of protesting against AI, which some consumers deemed futile. Thus, for some consumers, the negative attitude seemed to be overridden by the low perceived control, resulting in adoption intention.

Last, but not least, the results also point to the contextuality of consumer attitudes towards AI. While the prior research on consumer adoption and attitudes towards AI (e.g., Castillo et al. 2021; Hasan et al. 2021; Nagy and Hajdú 2021) was grounded in the market logic of the capitalist system, this article sheds new light on this subject by examining consumer behavior in the context of a counter-brand community (Cova and White 2010), where the co-created value is appropriated by the community members, rather than by the company. The research shows that the reasoning changes in this different context. This concerns reasons against adoption in particular, with consumers consciously dismissing ethical, legal, and social reasons against adoption due to the non-commercial character of AI mods, as well as the otherwise limited resources of mod authors. Moreover, while AI has often been associated by consumers with companies' attempts at cost-cutting in pursuit of profit, and the resulting diminished quality of the value proposition (Price 2023; Zhou 2023; Zhou and Kawabata 2023), in the case of mods, which are mostly free and usually created as a hobby, AI usage is seen by some consumers as more acceptable, and does not provoke an association with corporate greed. In more general terms, it can be argued that the negative perception of AI by consumers might not be exclusively tied to the nature of the technology itself, but may also be influenced by the context of the capitalist system, where said technology is being used to the detriment of consumers, workers, and society at large. At the same time, this perception can be more favorable in the case of alternative contexts, such as online communities, where value co-creation is not strictly motivated by profit.

8 | Discussion

8.1 | Relating the Findings to Earlier Works

While there are no studies focusing specifically on using AI in the context of value co-creation within an online community, the findings from this article are largely consistent with and elaborate on some of the recent literature examining AI from different perspectives and in the context of various sectors. First, the mixed attitudes of consumers towards AI observed in the netnographic study are reflected in multiple existing studies. For instance, according to Maduku et al. (2025), consumer adoption can be limited by technology anxiety. Similarly, Kamoopuri and Sengar (2024) observed a general avoidance of AI in some consumers, which also negatively impacts adoption. On the other hand, Schwing et al. (2025) noticed a positive impact of technology trust on consumer adoption. This study complements the earlier ones by examining both the positive and negative consumer attitudes towards AI and the broader technology at the same time. Moreover, by using qualitative methodology, the article was able to provide some deeper insights into the consumers' decision-making processes and identify more specific reasons for and against the adoption of AI.

As concerns the reasons for and against the adoption of AI and the related services, the findings support the claims of Bhardwaj et al. (2024) regarding the impact of ethical concerns and authenticity issues on consumer perception of AI. More specifically, Liu and Shi (2025) discuss how the data captured by AI

can be seen by consumers as exploitative, and how this impacts consumer adoption. This concern about the exploitative potential of AI is also present in the results of the netnographic research carried out in this article. The study also confirms the findings of Schwing et al. (2025), who claim that perceived usefulness positively influences consumer adoption, by showing that consumers' perception of the quality of AI-generated assets varies greatly, which then impacts their overall attitude towards the technology.

Finally, the study provides further empirical proof for the claims of Qi et al. (2024) and de Oliveira Santini et al. (2025), who point to the differences in AI acceptance depending on the product type, with AI being received less favorably in the case of hedonistic products (in comparison to utilitarian products). This is reflected in the ambivalent attitude towards AI displayed in the research carried out in this article, which focuses on hedonistic products. The study further builds on these earlier findings by showing that differences in terms of consumer attitude towards AI stem not only from the differences in the product category but also from the way they were produced, with consumers being more accepting towards AI utilized in the process of value co-creation by an online community. This finding can, in turn, be tied to and partially explained by the study by Zhang et al. (2025), which shows that AI adoption is positively impacted by the psychological ownership experienced by the consumers. This psychological ownership, developed in the process of value co-creation, can be another reason behind the increased consumer acceptance of AI within the context of an online community.

8.2 | Theoretical and Research Implications

The article makes several theoretical contributions to the existing consumer behavior literature (and BRT in particular), a summary of which can be found in Table A2. By strengthening quantitative methods with the netnography, underutilized in AI research, and focusing on a video game industry, which hasn't been examined in this context by the existing literature, the research provides an in-depth look at consumer attitudes towards AI and the resulting AI adoption. By doing so, the article responds to the call of Spais and Jain (2025) for multi-method research approaches to understanding consumer interactions in the AI era that "enhance findings and highlight research gaps in immersive technologies and their effects on consumer decision-making" (2025, 21). The article also responds to the call of Lim, Kumar, et al. (2023) for new research that explores consumer engagement with new-age technologies in terms of adoption and resistance, as well as its ethical and psychological dimensions. In particular, the research shows that the use of AI in modding, despite being controversial among video game consumers, does not diminish consumer adoption. However, the findings also demonstrate that high AI transparency does negatively impact consumer adoption, providing evidence for the existence of anti-AI bias in consumers, which is consistent with the existing literature (Chamberlain et al. 2018; Horton et al. 2023; Zhang and Gosline 2023). Moreover, by using the netnography the study highlights the consumers' ethical concerns regarding the use of AI. By doing so, the research also provides further support to the call of Spais and Jain (2025) for a holistic approach to Corporate

Digital Responsibility (CDR). The findings show that the unethical use of AI by companies is one of the reasons against adoption. Thus, CDR becomes not only a crucial ethical challenge but also a necessary response to the needs of the consumers.

By focusing on the online community built around the *Nexus Mods* platform, the study also responds to the call by Jain, Sharma, et al. (2024, 18) for studies exploring value co-creation "in emerging contexts like online brand communities, social media, and artificial intelligence." Simultaneously, the article addresses the gap identified by Beheshti et al. (2024) who state that the existing consumer-centric articles on disruptive technologies focus mostly on retail-related research, and thus suggest that future studies should investigate other contexts. In particular, the article makes a contribution to the AI adoption and online community literature by exploring the contextual-ity of consumer adoption and attitudes towards this technology. Whereas most studies focus on how consumers respond to AI in consumption that's largely controlled by companies (Castillo et al. 2021; Gaczek et al. 2023; Hasan et al. 2021; Nagy and Hajdú 2021), this research pays attention to the AI attitudes in an alternative, more consumer-driven environment of an online counter-brand community (Cova and White 2010) built through "participation from below" (Yeritsian 2021), which does not strictly adhere to the usual market logic of late capitalism. This provides additional insight regarding the consumers' attitude toward AI, showing that reasons for and against adoption differ depending on the context, with some consumers being more accepting towards AI when it's used by community members for joint value co-creation and appropriation, rather than by companies seeking financial gains.

8.3 | Managerial Implications

The research conducted in this article also carries a number of practical implications. First, AI is confirmed to be a controversial technology, which many consumers show bias against. This means that using AI as a marketing crutch, particularly in cases when it does not improve the actual quality of the value proposition or consumer experience (i.e., AI-washing) can, at least in some cases, become counterproductive and lead to consumers developing negative attitudes towards the company and its offering. Thus, a safer approach would be using AI sparingly, in a supplementary rather than substitutive role, as well as focusing the marketing communication on the actual benefits of the offering rather than the mere fact of it being based on AI technology.

For managers, the findings displaying the contextuality of consumer attitudes towards AI can also be valuable information. Consumers being seemingly more accepting of AI when it is not used as a cheap way to substitute human labor shows that they are aware of the broader impact of the technologies they use. Thus, companies wanting to implement AI-based solutions should also strive to develop more socially and ethically conscious versions of this technology, as well as its applications, to improve consumer attitude and adoption.

The contextual nature of consumer attitude towards AI can also be of interest to non-corporate organizations and communities,

who can potentially employ this technology while receiving less backlash than a company would. That being said, even when used in such alternative contexts, many ethical and legal problems inherent to the current AI models (such as being trained on unethically obtained data) remain. Similarly, the risk of misuse of AI also persists. Thus, even outside of the market logic of capitalism, the technology can lead to the development of negative consumer attitudes and diminish adoption. These problems could potentially be solved by establishing and enforcing more strict and detailed sets of rules regarding allowed AI usage within online communities. For instance, for content communities such as Nexus Mods, only ethically trained AIs could be allowed, and written permissions for imitating works and performances of concrete artists could be required. While this would most likely significantly increase the required workload in regard to content moderation, it would also help make the point that AI does not have to be inherently tantamount to late capitalist exploitation and establish a framework in which this technology could be used in order to support human creativity, rather than supplanting it.

8.4 | Limitations and Future Research

The research conducted in this article comes with some limitations. Despite including a large overall number of mods in the study, the number of mods using AI remains relatively low, limiting the potential sample size. This also did not allow for studying the differences in consumer responses between different applications of AI technology. The netnography did shed some light on this aspect, with AI imitating real actors' voices turning out to be particularly contentious (especially when used to voice mods containing adult content), but in general, the discussions on this subject were rare.

Another limitation of this study results from the lack of direct interviews with consumers, as the study used only non-participatory netnography. Thus, future studies could further investigate consumer attitudes towards AI in online community value co-creation via in-depth interviews with various members (such as users, modders, and moderators), which would provide a deeper understanding of reasons for and reasons against adoption, and their interplay, hierarchy, as well as their contextuality.

A more detailed analysis of what is and what is not acceptable in the eyes of consumers could also be achieved via further quantitative studies carried out once more mod creators start using AI. For instance, including additional variables such as mod category could provide more insights regarding the consumers' attitudes towards AI depending on its application.

Moreover, future research could try replicating this study in a different sector and focusing on different types of value co-creation in online communities. For instance, consumer attitudes toward the use of AI in fan-fiction communities would allow exploring consumer attitudes toward AI in a less tech-focused context.

To further elaborate on the contextuality of consumer attitudes toward AI, other contexts should also be explored. For instance,

the use of AI by non-profit organizations and the reactions of their supporters could be examined.

9 | Conclusion

The purpose of this article was to examine consumers' adoption of AI-based products co-created in online communities. Utilizing the theoretical framework of behavioral reasoning theory and online community literature, as well as the mixed-method study of the *Nexus Mods* community, the article provides answers to the four research questions formulated at the beginning of the article. In the context of RQ1 ("How does the use of AI in mod development impact consumer adoption?"), the findings of the quantitative study show that the use of AI in mod development not only does not diminish consumer adoption but actually improves it, albeit to a small degree. In response to RQ2 ("How does the AI transparency of AI mods impact consumer adoption?"), the quantitative results show that AI transparency negatively impacts consumer adoption of AI mods, pointing to the existence of anti-AI bias in consumers. As concerns RQ3 ("What are the reasons for and reasons against the consumer adoption of AI mods?"), the netnography identifies several reasons for the adoption of AI mods (high quality, enriched experiences, more choice, novelty of AI, consumer empowerment, consumer resistance) and against the adoption (low quality, "lack of soul", negative social impact, ethical considerations, legal concerns, and lack of transparency). As for RQ4 ("How does the context of the online community impact consumer adoption?"), the findings from netnography point to the importance of the online community context and its influence on reasons for and against the adoption of AI. In this alternative context, some consumers are more willing to accept AI, as it is used as a part of a hobby rather than for profit.

The study builds on and contributes to the consumer behavior and AI literature. It shows that despite its popularity, AI remains controversial among consumers, who, on the one hand, are willing to adopt AI mods, particularly when they aren't advertised as such, but, at the same time, still display anti-AI bias. Moreover, by revealing the contextual character of consumer attitudes, the study shows that the consumers' negative attitudes are tied more so to the way AI is utilized by companies rather than to the technology itself. Thus, both companies and governments should put more focus on the ethics of AI use and developing holistic CDR strategies.

Ethics Statement

This study was approved by the Committee of Ethical Science Research conducted with the participation of humans at PUEB (Resolution no. 17/2024) on August 19, 2024.

Consent

Informed consent for information published in this article was not obtained because it has been deemed that consent would be impossible or impracticable to obtain—this was approved by the Committee of Ethical Science Research conducted with the participation of humans at PUEB (Resolution no. 17/2024) on August 19, 2024.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

Due to the nature of the research and due to ethical/legal concerns, supporting data is not available.

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Appendix A

TABLE A1 | Reasons for and against AI mods adoption.

Reasons for AI mod adoption		Supporting quote(s)
High quality		“Stuff generated by Eleven Labs AI is almost indistinguishable from actual voice acting.” “To be honest, AI-generated art is amazing.”
Enriched experiences		“I’m really looking forward to seeing how this makes games feel more vibrant. This is exactly why people enjoy games like D&D. To experience that dynamic, living world. It won’t replace it, but it will definitely add to it.” “This introduces another dimension to modding, letting players dive deeper into the world of [game] by not just reading new content but actually hearing it. Although AI voices don’t sound completely lifelike, they work well enough for many.”
More choice		“The idea of creating endless variations and permutations of their work with just a keystroke? Bro, we’re on the brink of something incredible.”
Novelty		“This is a new tech. Yes, new things can be scary. Change can be scary, but if we don’t let this change happen, what might we miss out on by not allowing something as small as AI mods to spread its wings?” “I’m thrilled to see what the community will create with this tech!” “Adding AI-generated content will be a huge benefit for modding :)” “I hope we see a lot of good new content coming from this! Keep it going!” “Honestly, AI is where it’s at right now, so why not give it a try? It might help, it might not, or it could evolve into something even more useful for modders over time. We won’t know until we try ..”
Consumer empowerment		“AI is simply a tool that will speed up progress and make some mods, which were previously impossible, a reality (like adding dialogue to existing characters, quickly creating large levels, and assisting with writing dialogue).” “With these new tools, you (the unskilled individual reading this) are now more capable of creating mods than ever before. No special talent is needed, just a basic understanding of a game’s modding tools and a simple grasp of how to use an AI app.” “AI will enable more people to create things like voices and illustrations for their mods without having to spend ridiculous amount of time or money on something that’s just a hobby and for people who might not have the funds to begin with.” “People who aren’t skilled in art but are great at coming up with concepts can use AI to fill in the gaps, or creators who are already good at this can speed up or expand the scale of their mods.”
Consumer resistance		“Get rid of all copyright laws.” “If AI gets banned like some of you want, you’ll soon realize that all you’ve done is screw over the average Joe and give all the power to big corporations, who will use this technology and make sure they never lose it through lobbying.” “The ‘AI equals theft’ crowd doesn’t get that their stance could lead to big corporations having even more control over artists and their livelihoods.” “This divide between ‘hardworking, human, soulful artists’ and ‘evil, soulless, mechanistic thieves’ will only benefit those who want to solidify corporate ownership of art, giving control to those with money and titles rather than the actual artists who created it. When has that capitalist idea of copyright ever helped any artist?” “If AI gets banned and regulated, it will only be accessible to the government and big companies like Disney and Microsoft.”
Reasons against AI mod adoption		Supporting quote(s)
Low quality		“AI-generated stuff always looks uncanny to me, especially when it tries to create people. There’s always something off that makes me uncomfortable, and once I inevitably notice it, I can’t unsee it, making it unbearable.” “For me, AI is too easily recognizable and doesn’t sound good, it hurts my ears every time I hear it.” “Currently, AI-generated voices are far from replacing real voice actors in any significant project. There’s no emotion or insight, and it can’t ‘act’, it just repeats patterns.”

(Continues)

TABLE A1 | (Continued)

Reasons against AI mod adoption	Supporting quote(s)
"Lack of soul"	"Outsourcing things like quest design, art assets, and environment development to a machine is an insult to those who create these amazing mods in their free time. Anyone supporting AI likely never spent any time creating something like a short story, a piece of art, or music." "Only humans can create art because it's made with the soul, not the hands or mouth. Everything else, no matter how advanced, is just a tool." "Frankly, I really dislike calling AI-generated content 'art'. To me, art is about the artist's expression. AI art is just the output of a soulless, cold algorithm."
Negative social impact	"They will use AI to speed up production, but the trade-off is losing genuine passion and sometimes real jobs." "It's not even better than most mid-level artists; it's just 'good enough' for them to feel comfortable replacing them. Pushing out everyone except the top-tier artists isn't beneficial."
Ethical considerations	"I'd be more concerned about the ethics and data privacy issues." "It's disappointing that voice actors aren't getting more protection on this website, especially when AI is being used for voicing NSFW [not safe for work; containing explicit content] mods." "What really frustrates me is that its main use right now is turning women to sex objects." "Voice actors shouldn't have to constantly monitor mods to ensure their voices aren't being stolen. Nexus's policy seems to imply that only the voice actor can file a complaint to get the mod taken down."
Legality concerns	"If the AI is being used for profit, I can't see the law allowing the training data to steal copyrighted content without permission to create and sell a product." "Describing it as 'AI-generated' is intentionally misleading and pretentious. It's much closer to algorithm-assisted stealing."
Lack of transparency	"This will be a problem to Nexus users if this spreads, forcing them to sort through bot-made art and human-made art. New tech isn't always good."
Behavioral context	Supporting quote(s)
Online community	"Modding games doesn't significantly impact voice actors in the grand scheme of things. The real issue will be when video game and cartoon studios start replacing voice actors with AI. [...] Some modding teams might have budgets, but most mods are solo projects done as a hobby. I don't think mod makers have an obligation to hire people." "It's not that there are no issues with AI or how it's currently being used and abused by capitalist system, but it's not like the art will be dead either. Most modders don't ask for money and only share their mods with people who use them on their private PCs in their free time. This isn't exactly making a fortune off of plagiarism." "Modding is about modifying things—this isn't new, but for some reason, people are acting like it is. We're not the ones threatening the livelihoods of the original artists, it's the industry."

TABLE A2 | Literature review and contributions.

Prior research	Contributions of this study
de Bellis and Venkataramani Johar (2020); Kumar et al. (2024); Guo et al. (2023); Spais et al. (2024); Huang and Qian (2021); Lee and Chen (2022): Adoption of AI-based services and products	The study extends the research on AI adoption to the unexplored video game industry.
Maduku et al. (2025); Kamoopuri and Sengar (2024); Schwing et al. (2025): Positive and negative consumer attitudes towards AI and technology	The study explores mixed attitudes toward AI technology within the same context of an online community.
Sung et al. (2021): AI quality and novelty as factors influencing consumer adoption	The study confirms that both the low and high quality can become a reason against or for adoption, respectively. Similarly, novelty has been confirmed as a potential reason for adoption.
Han and Broniarczyk (2022): AI hypothesized as the source of new, rich experiences and consumer empowerment	The study provides empirical evidence for the claim based on the netnography.
Bhardwaj et al. (2024); Gerlich (2023): Ethical considerations and societal impact influence consumer acceptance of AI	The study provides empirical support for the claim based on the netnography.
Bellaiche et al. (2023); Zhou and Kawabata (2023): AI being seen by consumers as lacking creativity, unable to reflect profound human experience, and harder to engage deeply or derive meaning from	The study strengthens this claim by showing that AI creations and mods are perceived as “soulless” by consumers.
Chamberlain et al. (2018); Horton et al. (2023): The existence of anti-AI bias in consumers in regard to AI-generated art	The study demonstrates the existence of anti-AI bias against products developed by an online community.
Luo et al. (2019): The impact of AI transparency on consumer behavior	The study examines the impact of AI transparency on consumer adoption within the online community.
Castillo et al. (2021); Hasan et al. (2021); Nagy and Hajdú (2021): The consumer attitudes towards AI in the context of market logic	The study shows the impact of community context on consumer attitudes towards AI.
Cova and White (2010): Counter-brands conceptualized as online communities that resist companies	The study explores AI as a tool for consumer resistance against companies within an online community. It also further examines the adoption of community-developed products.
Qi et al. (2024); de Oliveira Santini et al. (2025): The dependence of AI acceptance on the product type (hedonistic vs. utilitarian)	The study shows that AI acceptance is mixed in the case of hedonistic products.
Zhang et al. (2025): The impact of psychological ownership on AI adoption	The study demonstrates that consumers are more accepting of AI in the context of online community value co-creation, which involves higher psychological ownership (in comparison to the context of the typical market).

Source: Own elaboration.